Mitchell Knoth

Cloud Software Engineer | Developer platforms and coding-agent infrastructure

Des Moines, Iowa | mknoth197@gmail.com | Portfolio | GitHub | LinkedIn

Developer-platform engineer building reliable coding-agent execution, shared context, and production engineering-data systems. Cloud Software Engineer at John Deere since 2019, with experience in backend reliability, cloud networking, and secure delivery.

EXPERIENCE - John Deere | 2019 - present

Cloud Software Engineer - selected areas of responsibility

AI-assisted delivery and developer platforms | 2026 - present

- Wrote product requirements and much of the design record; delivered live-data views and a personal work timeline for a team-owned engineering evidence product now in production.
- Implemented Redis caching with tier-specific lifetimes and source-update watermarks; cache failures bypass to the source, and source errors remain visible.
- Separated CI validation from deployment and contributed dedicated acceptance checks for engineering views.
- Designed a shared knowledge repository with hybrid retrieval, a structural graph, a memory lifecycle, and CI lint checks; co-facilitate a guild and office hours for around 38 practitioners.
- Ran Codex and Claude Code in disposable cloud sandboxes and implemented repository-credential proxying; returned logs and inspectable patches. Separate keyless pilot is merged, with live verification incomplete.
- Built a versioned skill-scanning trust layer with separate policy decisions and SARIF output; packaged reusable pre-PR reviews and feedback hooks.

Developer Experience | Late 2025 - early 2026

- Hardened the separation-of-duties platform with Datadog monitors, dashboards, and synthetics; added automatic incidents for pipeline failures and consolidated a legacy processor.
- Worked through CVE and Security Hub findings, migrated vulnerability-data integrations, reviewed organization governance, and introduced agent tooling into team repositories.

Enterprise cloud foundations | 2023 - 2025

- Led NAT Gateway consolidation, orchestrated a DNS migration, and automated certificate validation and renewal paths.
- Replaced manual account and VPC deletion with Step Functions and Lambda workflows; built DNS verification and cleanup and introduced reproducible Dev Container environments.

Digital equipment licensing | 2021 - 2023

- During a hackathon, built a secure license-data ingestion prototype that kept isolation and access controls intact; the prototype helped make the case for an enterprise Data Engineering team.

External API platform | 2019 - 2021

- Architected the platform’s first external-facing dealer API with role-based authorization and audit logging; the design became a reusable pattern for external access.

PUBLIC IMPLEMENTATION

Bounded execution lab - Independent Docker experiment with synthetic credential mediation, returned patches, and separate validation; includes recorded accepted, rejected, and timed-out fixture runs.

TECHNICAL FOCUS

TypeScript, Python, Java; AWS, Terraform, Step Functions, Lambda; Redis; GitHub Actions; Datadog; identity and credential boundaries; agent harness and context engineering.

EDUCATION & CERTIFICATIONS

B.S. Software Engineering, Iowa State University, 2020. Cum Laude; cyber security minor.

AWS Certified DevOps Engineer - Professional (May 2026) · AWS Certified Advanced Networking - Specialty (August 2025) · AWS Certified SysOps Administrator - Associate (February 2025) · AWS Certified Solutions Architect - Associate (March 2024) · AWS Certified Cloud Practitioner (November 2021)